

Route C Bootstrap Derivation Ledger through P16-V

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Abstract

This note records the derivations generated in Route C stages P1–P16-V. It is a derivation ledger, not a claim that on-shell consistency has already derived a complete GUT. The current Route C program uses the one-family Standard Model spectrum with a right-handed neutrino as baseline, builds charge-filtered symbolic four-point pole components, checks conditional residue factorization/positivity, marks Ward-identity completion requirements, compares anomaly/chiral consistency across $SU(5)$, Pati–Salam, $Spin(10)$, E_6 , and superstring-derived completions, and classifies controlled high-energy-growth and low-energy proton-bound readiness requirements, and closes the first pass with a comparative theory scorecard. P9 then locks the branch to $Spin(10)$ and supplies the first sparse action-completion ledger for the half-spinor broken-generator maps and the remaining Goldstone/Higgs, mediator, flavor, and proton-bound inputs. P10 chooses a staged source branch and converts the symbolic vector masses into a diagonalizable mass matrix with one hypercharge zero mode. P11 replays the P6/P7 matching gates with those staged masses and separates adjoint current-vector matching from scalar/source chiral-pair mass debt. P12-S records both post-P11 directions, activates the scalar/source chiral-pair branch first, and assigns symbolic scalar/source mass denominators to all P7 chiral-pair rows while leaving physical proton bounds conditional. P13-S chooses a symbolic scalar/source flavor-tensor basis and maps the BNV rows to an all-left chiral Wilson basis, while retaining the Fierz/color recoupling and physical-flavor debts. P14-V returns to the parallel $Spin(10)$ current-current vector branch and builds the vector matching gate before any proton-limit insertion. P15-V fixes the first two-component Fierz/crossing convention for that vector branch and audits the crossed all-left skeletons. P16-V assembles physical Hermitian cross-root vector multiplets and finds the first canonical vector BNV skeletons in the off-face $M_{(6,2,2)}$ block, while leaving Clebsch phases, flavor, RG, hadronic, and proton-limit inputs conditional.

1 Boundary of the Ledger

Route C tests whether

$$\begin{aligned} & \text{Lorentz} + \text{unitarity} + \text{factorization} + \text{Ward identities} \\ & + \text{anomaly cancellation} + \text{controlled high-energy behavior} \end{aligned} \tag{1}$$

can filter conditional GUT branches. It must not be read as

$$\text{four axioms} \implies \text{the full GUT.} \tag{2}$$

The honest statement is instead

$$\text{on-shell consistency} \implies \text{filters on allowed action-level GUT branches.} \tag{3}$$

The baseline external spectrum is one left-handed family plus a right-handed-neutrino conjugate field,

$$\mathcal{F}_{\text{SM}+\nu^c} = Q \oplus L \oplus u^c \oplus d^c \oplus \nu^c \oplus e^c. \tag{4}$$

All fields are written as left-handed Weyl fields.

2 P1: External-State Charges and Anomalies

2.1 Charge Table

The P1 charge ledger fixes the Standard Model face conventions shown in Table 1.

state	$SU(3)_C$	$SU(2)_L$	Y	$B - L$	multiplicity
Q	3	2	1/6	1/3	6
L	1	2	-1/2	-1	2
u^c	$\bar{3}$	1	-2/3	-1/3	3
d^c	$\bar{3}$	1	1/3	-1/3	3
ν^c	1	1	0	1	1
e^c	1	1	1	1	1

Table 1: P1 baseline spectrum. The fields u^c, d^c, ν^c, e^c are left-handed conjugate Weyl fields, so the listed charges are the charges of those conjugate fields.

2.2 Hypercharge Normalization

For one full family plus ν^c , the hypercharge trace is

$$\begin{aligned} \text{Tr} Y^2 &= 6 \left(\frac{1}{6} \right)^2 + 2 \left(-\frac{1}{2} \right)^2 + 3 \left(-\frac{2}{3} \right)^2 + 3 \left(\frac{1}{3} \right)^2 + 1 \cdot 0^2 + 1 \cdot 1^2 \\ &= \frac{10}{3}. \end{aligned} \tag{5}$$

With $\text{Tr}_2 T_{3L}^2 = 1/2$, the weak trace is

$$\text{Tr} T_{3L}^2 = 3 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = 2. \tag{6}$$

Therefore

$$\frac{\text{Tr} Y^2}{\text{Tr} T_{3L}^2} = \frac{5}{3}, \tag{7}$$

which is the standard GUT hypercharge normalization.

2.3 Perturbative and Global Anomaly Checks

With the conventions of Table 1, the exact rational ledger verifies

$$\mathcal{A}_{SU(3)^3} = 2A(3) + A(\bar{3}) + A(\bar{3}) = 2 - 1 - 1 = 0. \tag{8}$$

The mixed color-hypercharge anomaly is

$$\mathcal{A}_{SU(3)^2 U(1)_Y} = 2T(3)\frac{1}{6} + T(\bar{3})\left(-\frac{2}{3}\right) + T(\bar{3})\frac{1}{3} = \frac{1}{6} - \frac{1}{3} + \frac{1}{6} = 0, \tag{9}$$

where $T(3) = T(\bar{3}) = 1/2$. The weak-hypercharge anomaly is

$$\mathcal{A}_{SU(2)^2 U(1)_Y} = 3T(2)\frac{1}{6} + T(2)\left(-\frac{1}{2}\right) = \frac{1}{4} - \frac{1}{4} = 0. \tag{10}$$

The gravitational and cubic hypercharge anomalies are

$$\mathcal{A}_{\text{grav}^2 U(1)_Y} = 6\frac{1}{6} + 2\left(-\frac{1}{2}\right) + 3\left(-\frac{2}{3}\right) + 3\frac{1}{3} + 0 + 1 = 0, \quad (11)$$

$$\mathcal{A}_{U(1)_Y^3} = 6\left(\frac{1}{6}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 0^3 + 1^3 = 0. \quad (12)$$

The same exact ledger verifies the analogous $B-L$ and mixed $Y-B-L$ anomalies,

$$\text{SU}(3)^2(B-L), \quad \text{SU}(2)^2(B-L), \quad \text{grav}^2(B-L), \quad (B-L)^3, \quad Y^2(B-L), \quad Y(B-L)^2, \quad (13)$$

all vanish. The Witten $\text{SU}(2)$ check passes because one family contains three Q doublets and one L doublet, hence four left-handed doublets.

3 P2: Symbolic Four-Point Pole Ansatz

The first pole ansatz is an all-incoming chiral-bilinear seed,

$$\mathcal{A}_4(12|34) = \sum_X \frac{R_s(X)}{s - M_X^2} + C_6(12|34), \quad R_s(X) = g_{12X} g_{34\bar{X}}. \quad (14)$$

P2 does not declare the mediator spectrum. It only computes which symbolic pole components are not forbidden by the P1 charges and by non-Abelian conjugacy of pair products.

The generated P2 ledger gives the numerical summary

quantity	value	
pair types	21	
pair-product components	31	
tested pair-component pairings	496	(15)
allowed symbolic pole components	18	
forbidden by Abelian charge	455	
forbidden after Abelian pass by non-Abelian conjugacy	23	

The proton-operator seeds retained at this purely symbolic level are

$$QQQL : 2 \text{ allowed pole components}, \quad u^c u^c d^c e^c : 2 \text{ allowed pole components}. \quad (16)$$

The entries

$$QQu^c e^c, \quad QLu^c d^c \quad (17)$$

have zero allowed pole components in this first all-incoming symbolic seed. These zeros are not no-go theorems for full models; they are bookkeeping results for the declared P2 ansatz and charge conventions.

4 P3: Residue Factorization and Positivity

For a declared mediator sector X , let α label all two-particle channels that can couple to X . Near a positive-norm simple pole, tree-level factorization requires

$$\text{Res}_X \mathcal{A}_{\alpha\beta} = g_{\alpha X} g_{\beta X}^*. \quad (18)$$

Thus the residue matrix is a Gram matrix,

$$R_X = g_X g_X^\dagger, \quad (19)$$

and is positive semidefinite in a positive-metric Hilbert space:

$$v^\dagger R_X v = |g_X^\dagger v|^2 \geq 0. \quad (20)$$

A wrong-sign mediator metric reverses the sign and fails the unit-coupling stress test.

The P3 machine ledger checks the P2-allowed poles and reports

quantity	value
mediator sectors	26
total vertex channels	31
P2 allowed poles checked	18
factorization checks passed	18
PSD blocks under positive metric	26
largest channel-block size	3

(21)

For a unit-coupling block with n channels, the stress spectrum is

$$\text{spec}(R_X) = \{n, 0, \dots, 0\}, \quad \text{spec}(-R_X) = \{-n, 0, \dots, 0\}, \quad (22)$$

which is the numerical pattern seen in the P3 output.

5 P4: Ward-Identity Bookkeeping

For an unbroken massless gauge boson, the target Ward check is

$$\epsilon_\mu \rightarrow p_\mu, \quad p_\mu \mathcal{M}^\mu = 0. \quad (23)$$

For a broken massive vector, the target relation is the generalized Ward/Slavnov–Taylor form

$$p_\mu \mathcal{M}^\mu(V_X) = M_X \mathcal{M}(\phi_X), \quad (24)$$

so a Goldstone/Higgs/source completion is required before a true Ward identity proof can be claimed.

The P4 ledger separates unbroken SM currents from broken/off-face currents:

quantity	value
unbroken SM current sectors	3
current transitions checked	36
massless diagonal transitions ready	6
broken/off-face transitions needing completion	30
mediator sectors inspected	26
mediator sectors needing completion if vector	26

(25)

The bookkeeping-ready unbroken sectors are

$$\text{SU}(3)_C, \quad \text{SU}(2)_L, \quad U(1)_Y. \quad (26)$$

All off-face Spin(10), Pati–Salam leptoquark, and $\text{SU}(2)_R$ charged-current transitions are flagged as needing explicit generator matrices, massive-vector data, and Goldstone/Higgs/source completion.

6 P5: Anomaly and Chiral-Consistency Candidate Ledger

P5 compares candidate branches against the P1 baseline. The structural filter used for the minimal Route C field-theory comparison is

$$\begin{aligned} &\text{simple group} + \text{one irreducible family object} + \text{contains } \nu^c \\ &+ \text{no chiral SM exotics} + \text{minimal standard realization.} \end{aligned} \quad (27)$$

6.1 SU(5)

One family plus ν^c is realized as

$$10 \oplus \bar{5} \oplus 1. \quad (28)$$

The cubic anomaly cancels:

$$A(10) + A(\bar{5}) + A(1) = 1 - 1 + 0 = 0. \quad (29)$$

Also $\pi_4(\text{SU}(5)) = 0$ in the standard four-dimensional global-anomaly check. However, the family is not one irreducible object; therefore SU(5) is anomaly-consistent but fails the single-object filter.

6.2 Pati–Salam

The standard Pati–Salam family is

$$(4, 2, 1) \oplus (\bar{4}, 1, 2) \quad \text{under} \quad \text{SU}(4)_C \times \text{SU}(2)_L \times \text{SU}(2)_R. \quad (30)$$

The $\text{SU}(4)_C^3$ anomaly cancels because

$$2A(4) + 2A(\bar{4}) = 2 - 2 = 0. \quad (31)$$

The Witten checks pass because $\dim(4) = 4$ gives four $\text{SU}(2)_L$ doublets and four $\text{SU}(2)_R$ doublets. Pati–Salam is physically natural and anomaly-consistent, but it is a product group and not a single irreducible family object under a simple group.

6.3 Spin(10)

The standard Spin(10) family is the half-spinor

$$16. \quad (32)$$

It contains the full

$$Q \oplus L \oplus u^c \oplus d^c \oplus \nu^c \oplus e^c \quad (33)$$

after restriction to the Standard Model face. The D_5 algebra has no independent cubic gauge-anomaly invariant in this standard check, and $\pi_4(\text{Spin}(10)) = 0$. Therefore Spin(10) : 16 is anomaly-safe and is the standard minimal single-object survivor under the stated Route C pre-filter.

6.4 E6

The E_6 fundamental representation restricts as

$$27 \longrightarrow 16 \oplus 10 \oplus 1 \quad \text{under Spin}(10). \quad (34)$$

It contains a Spin(10) family, but also contains extra $10 \oplus 1$ states. Since E_6 has no independent cubic gauge-anomaly invariant for the standard 27, the anomaly status is acceptable, but the branch requires an explicit lifting/vectorlike audit for the extra states. It is therefore not the minimal Route C survivor.

6.5 Superstring-Derived Completions

Superstring-derived completions are not a single four-dimensional field-theory group. They form a UV-completion class. A string-derived branch must supply compactification-specific data:

$$\text{Green-Schwarz/tadpole/modular consistency,} \quad \text{massless spectrum,} \quad \text{chiral index,} \quad (35)$$

together with exotics lifting and matching to the Route-A representation/index skeleton. Such branches may provide stronger high-energy softness or tower structure, but P5 does not treat them as a single fixed four-dimensional candidate.

6.6 P5 Numerical Summary

The generated P5 ledger reports

quantity	value
candidates inspected	5
field-theory candidates	4
UV-completion classes	1
P1 baseline anomalies pass	true
minimal single-object survivor	Spin(10)

(36)

This result should be read narrowly: passing anomaly checks alone does not select a GUT. The selection of Spin(10) occurs only after imposing the Route C minimal single-object family filter.

7 P6: Controlled High-Energy-Growth Audit

P6 consumes the P3 residue blocks and the P4 Ward-completion ledger. Its purpose is not to compute a full four-point amplitude. Its purpose is to classify which symbolic branches would have uncontrolled high-energy growth if treated as isolated finite broken-vector exchange.

For a finite massive vector, the dangerous contribution is the longitudinal propagator term. A conservative schematic form is

$$\mathcal{A}_L \sim \frac{(J \cdot p)(J' \cdot p)}{M_X^2(s - M_X^2)}. \quad (37)$$

If the broken current is not completed by the corresponding Goldstone/Higgs sector, this can scale as

$$\mathcal{A}_L \sim \frac{s}{M_X^2} \quad \text{or worse} \quad (38)$$

in the high-energy regime. Therefore the required broken-vector target is again the generalized Ward/Slavnov-Taylor identity

$$p_\mu \mathcal{M}^\mu(V_X) = M_X \mathcal{M}(\phi_X). \quad (39)$$

The generated P6 ledger reports

quantity	value
mediator sectors inspected	26
finite-vector branches needing completion	26
unbroken-vector branches bookkeeping-ready	0
scalar/auxiliary branches without longitudinal-vector risk	26
contact-only components needing UV completion	18
broken transitions needing completion	30
tower-like UV forced now	false

(40)

Thus every current P3 mediator sector, if interpreted as a finite broken vector, needs explicit finite-completion data:

$$\{\text{broken generator, } M_X, \phi_X, \text{Higgs/source sector, symmetry-fixed contact/numerator terms}\}. \quad (41)$$

If a symbolic mediator is instead interpreted as scalar or auxiliary exchange, the longitudinal-vector Ward problem is absent. This is only a bookkeeping statement; spin-statistics, action existence, low-energy matching, and perturbative unitarity remain separate checks.

A contact-only four-fermion EFT normalization generically scales like

$$\mathcal{A}_{\text{contact}} \sim \frac{s}{\Lambda^2}, \quad (42)$$

so contact terms may encode low-energy matching but cannot by themselves serve as high-energy softness proofs.

Finally, P6 does not force a string or tower completion. It states the conditional comparison rule:

$$\begin{array}{l} \text{finite Goldstone/Higgs/source completion fails} \\ \text{tower-like UV completion becomes a serious comparison class.} \end{array} \implies \quad (43)$$

Superstring-derived completions therefore remain comparison branches, not P6 conclusions.

8 P7: Low-Energy Matching and Proton-Bound Readiness

P7 converts the P2/P3 symbolic pole data into conditional low-energy dimension-six matching rows. It respects the P6 gate: a vector interpretation of a broken mediator cannot be promoted to a physical Wilson coefficient until the finite Goldstone/Higgs/source completion has been supplied.

For a symbolic pole with mediator X , P7 records the matching template

$$C_6[\mathcal{O}; X] = \frac{g_{\text{left},X} g_{\text{right},X}^*}{M_X^2}. \quad (44)$$

This is only a gauge-basis symbolic coefficient. It is not yet a physical proton-decay Wilson coefficient.

Using the left-handed Weyl charge convention, the global bookkeeping assigns

$$B(Q) = \frac{1}{3}, \quad B(u^c) = B(d^c) = -\frac{1}{3}, \quad L(L) = 1, \quad L(\nu^c) = L(e^c) = -1. \quad (45)$$

The generated P7 ledger reports

quantity	value
allowed symbolic poles processed	18
B and L conserving poles	12
standard BNV seed poles	4
BNV-with-sterile-neutrino poles	2
baryon-violating symbolic poles	6
BNV vector poles blocked by P6 completion	6
physical proton bounds evaluable now	0
all physical bounds conditional	true

(46)

The standard baryon-violating seeds include

$$QQQL, \quad u^c u^c d^c e^c, \quad (47)$$

and the sterile-neutrino branch includes

$$u^c d^c d^c \nu^c. \quad (48)$$

All of these preserve $B - L$ in the P1 bookkeeping.

The physical proton-bound interface is recorded as

$$\Gamma(p \rightarrow a) = \sum_{i,j} C_i^{\text{phys}} C_j^{\text{phys}*} H_{ij}^{(a)}, \quad (49)$$

where $H_{ij}^{(a)}$ contains phase space, RG evolution, chiral reduction, and hadronic matrix elements for channel a . A lifetime lower bound implies

$$(C^{\text{phys}})^\dagger H^{(a)} C^{\text{phys}} < \frac{1}{\tau_a^{\text{exp}}}. \quad (50)$$

For a single coefficient this reduces schematically to

$$|C_i^{\text{phys}}| < \sqrt{\frac{1}{\tau_a^{\text{exp}} H_{ii}^{(a)}}}. \quad (51)$$

P7 deliberately does not insert numerical proton limits. Numerical lifetimes require:

$$\begin{aligned} &\text{P6-completed high-energy branch,} \quad \text{mediator masses and couplings,} \quad \text{physical flavor rotations,} \\ &\text{operator basis and chiral contractions,} \quad \text{RG factors,} \quad \text{lattice/chiral matrix elements,} \quad \tau_a^{\text{exp}}. \end{aligned} \quad (52)$$

Thus P7 is a proton-bound readiness report, not a proton-lifetime calculation.

9 P8: Comparative Theory Scorecard

P8 compares candidate branches using the explicit P1–P7 gates. It is a hard-gate ledger, not a subjective weighted score. The gates are:

C0	representation pre-filter
P1	charge/anomaly baseline
P2	charge-filtered pole ledger
P3	residue factorization/PSD
P4	Ward-completion readiness
P5	anomaly/chiral/exotics ledger
P6	high-energy-growth completion gate
P7	proton-bound readiness

The generated P8 scorecard compares five candidates. Its leading conditional finite field-theory branch is $\text{Spin}(10)$, and it does not yet force a tower-like UV completion. The candidate verdicts are:

candidate	P8 verdict
SU(5)	conditional branch if the single-object filter is relaxed; anomaly-consistent but family-split as $10 \oplus \bar{5} \oplus 1$
Pati–Salam	conditional intermediate face; natural and anomaly-consistent, but product group and not one simple single-object branch
Spin(10)	leading conditional finite field-theory branch under the current minimal single-object filter; still has action-level and proton-bound completion debt
E_6 superstring-derived completions	conditional branch with exotics-lifting debt from $27 \rightarrow 16 \oplus 10 \oplus 1$ UV-completion class, not forced by P1–P7; serious comparison branch if finite-pole completions fail

Thus P8 closes the first Route C pass as a comparative filter:

$$\text{Spin}(10) = \text{leading conditional finite branch}, \quad \text{not a completed GUT proof.} \quad (53)$$

Superstring-derived completions remain comparison branches, not P8 conclusions.

10 P9: Spin(10)-Locked Action-Completion Data

P9 stops broadening the candidate comparison and locks the finite field-theory branch to the P8-leading Spin(10) option. The generated ledger uses the Pati–Salam decomposition

$$16 \longrightarrow (4, 2, 1) \oplus (\bar{4}, 1, 2), \quad (54)$$

and builds a sparse sixteen-state basis for

$$Q \oplus L \oplus u^c \oplus d^c \oplus \nu^c \oplus e^c. \quad (55)$$

The broken-generator data are represented as sparse transition maps rather than as a completed Lagrangian. P9 reports:

quantity	value
basis dimension	16
broken generator sparse maps	32
SU(4) _C leptoquark root maps	6
broken SU(2) _R charged root maps	2
Spin(10)/Pati–Salam (6, 2, 2) Clebsch maps	24
P4 broken transition pairs	30
Spin(10) adjoint vector transition pairs	22
non-adjoint P4 completion pairs	8
P9 generator layer passes	true

The important refinement is that not every P4 charge-bookkeeping transition is a Spin(10) adjoint gauge vector. The eight non-adjoint P4 pairs are not generator failures; they remain scalar, source, or auxiliary completion candidates if a later branch activates them. P9 therefore supplies the generator layer without pretending that charge-allowed transitions are automatically D_5 gauge generators.

For every broken vector, the action-level Ward target remains

$$p_\mu \mathcal{M}^\mu(V_X) = M_X \mathcal{M}(\phi_X). \quad (56)$$

P9 records the corresponding completion templates for

$$\text{Spin}(10) \rightarrow \text{Pati-Salam}, \quad \text{SU}(4)_C \rightarrow \text{SU}(3)_C \times U(1)_{B-L}, \quad \text{SU}(2)_R \rightarrow U(1)_{T_{3R}}. \quad (57)$$

It also records the still-symbolic inputs

$$g_{10}, \quad M_{(6,2,2)}, \quad M_{\text{LQ}}, \quad M_{W_R}, \quad (58)$$

plus the physical flavor rotations and proton-bound interface. Thus P9 is an action-completion interface, not yet a spectrum choice, flavor fit, or proton-lifetime calculation.

11 P10: Spin(10) Breaking Mass Matrix

P10 chooses the conservative staged branch

$$\begin{aligned} \text{Spin}(10) &\rightarrow \text{SU}(4)_C \times \text{SU}(2)_L \times \text{SU}(2)_R, \\ \text{SU}(4)_C &\rightarrow \text{SU}(3)_C \times U(1)_{B-L}, \quad \text{SU}(2)_R \rightarrow U(1)_{T_{3R}}, \\ U(1)_{T_{3R}} \times U(1)_{B-L} &\rightarrow U(1)_Y. \end{aligned} \quad (59)$$

The branch is a source-level mass model, not a complete Higgs potential. It is chosen because it turns the P9 symbolic inputs into positive blocks while preserving the hypercharge zero mode.

The resulting active vector mass matrix has the block form

$$M_{\text{active}}^2 = \text{diag} \left(M_{(6,2,2)}^2 I_{24}, M_{\text{LQ}}^2 I_6, M_{W_R}^2 I_2, M_{\text{neutral}}^2 \right), \quad (60)$$

where

$$M_{(6,2,2)}^2 = \kappa_{\text{PS}} g_{10}^2 v_{\text{PS}}^2, \quad M_{\text{LQ}}^2 = \frac{2}{3} g_4^2 v_4^2, \quad M_{W_R}^2 = g_R^2 v_R^2. \quad (61)$$

The leptoquark coefficient follows from the normalized generator

$$T_{B-L} = \sqrt{\frac{3}{8}} \text{diag} \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1 \right), \quad (\Delta T_{B-L})^2 = \frac{2}{3}. \quad (62)$$

For the neutral sector, a $Y = 0$ ν^c -like source with charges $q_R = -1/2$ and $q_{B-L}/2 = +1/2$ gives

$$M_{\text{neutral}}^2 = \frac{v_Y^2}{4} \begin{pmatrix} g_R^2 & -g_R g_{BL} \\ -g_R g_{BL} & g_{BL}^2 \end{pmatrix}_{(W_R^3, B_{B-L})}. \quad (63)$$

This matrix has eigenvalues

$$0, \quad \frac{v_Y^2}{4} (g_R^2 + g_{BL}^2), \quad (64)$$

with massless hypercharge direction

$$B_Y \propto g_{BL} W_R^3 + g_R B_{B-L}, \quad g_Y = \frac{g_R g_{BL}}{\sqrt{g_R^2 + g_{BL}^2}}. \quad (65)$$

In the unit demo point

$$g_{10} = g_4 = g_R = g_{BL} = v_{\text{PS}} = v_4 = v_R = v_Y = 1, \quad \kappa_{\text{PS}} = 1, \quad (66)$$

P10 finds a 34-dimensional active matrix, 33 positive broken eigenvalues, and one zero eigenvalue. This is the expected count for

$$\dim \text{Spin}(10) - \dim G_{\text{SM}} = 45 - 12 = 33. \quad (67)$$

The remaining physical inputs are the scale choices, scalar/source branch for the eight non-adjoint P4 pairs if activated, physical flavor rotations, RG factors, hadronic matrix elements, and experimental proton lifetime limits.

12 P11: Replay with Staged Spin(10) Masses

P11 attaches the P10 mass blocks to the matching gates:

$$\begin{aligned}
M_{(6,2,2)} &\rightarrow \sqrt{\kappa_{\text{PS}}} g_{10} v_{\text{PS}}, \\
M_{\text{LQ}} &\rightarrow \sqrt{\frac{2}{3}} g_4 v_4, & M_{W_R} &\rightarrow g_R v_R, \\
M_{Z'} &\rightarrow \frac{1}{2} v_Y \sqrt{g_R^2 + g_{BL}^2}.
\end{aligned} \tag{68}$$

The current-vector side is therefore improved: the adjoint-vector sectors now carry explicit staged denominators. However, P11 also checks the P7 chiral-pair pole rows against the adjoint-current charge signatures. The result is:

quantity	value
current-vector mass blocks available	4
P7 rows replayed	18
P7 rows with candidate adjoint mass blocks	0
P7 rows remaining scalar/source mass debt	18
baryon-violating or sterile-BNV rows	6
physical proton bounds evaluable now	0

This is a negative but useful gate: the P7 all-incoming chiral-pair pole ledger is not automatically the same as adjoint current-vector exchange. Thus P11 does not unlock proton bounds. It says that the next Wilson-basis choice is unavoidable:

$$\text{current-current vector branch} \quad \text{or} \quad \text{scalar/source chiral-pair branch.} \tag{69}$$

The two branches require different spin/channel maps before numerical proton limits can be inserted.

13 P12-S: Scalar/Source Chiral-Pair Branch

P12 records both post-P11 directions:

$$\text{Branch V} = \text{completed Spin(10) current-current vector exchange,} \tag{70}$$

and

$$\text{Branch S} = \text{scalar/source chiral-pair poles.} \tag{71}$$

Both directions remain part of the Route C program. The first activated direction is Branch S, because P11 found that all P7 chiral-pair rows remain scalar/source mass debt rather than adjoint-vector matches.

The scalar/source ansatz is

$$\mathcal{L} \supset \lambda_{ab,R} \psi_a \psi_b S_R + \text{h.c.} + M_{S_R}^2 |S_R|^2, \quad M_{S_R}^2 > 0. \tag{72}$$

Integrating out the scalar/source field at tree level gives the symbolic dimension-six coefficient

$$C_6(12|34; R) = \frac{\lambda_{12,R} \lambda_{34,R}^*}{M_{S_R}^2}. \tag{73}$$

This avoids the longitudinal broken-vector Ward problem, but it does not prove a UV completion or a physical proton lifetime.

The generated P12-S ledger reports:

quantity	value
P11 scalar/source debt rows converted	18
unique scalar/source sectors	12
B and L conserving rows	12
standard BNV seed rows	4
BNV-with-sterile-neutrino rows	2
baryon-violating or sterile-BNV rows	6
physical proton bounds evaluable now	0

The six baryon-violating or sterile-BNV scalar/source rows are the two $QQQL$ rows, the two $u^c u^c d^c e^c$ rows, and the two $u^c d^c d^c \nu^c$ rows. Their coefficients remain symbolic because P12-S has not chosen the scalar/source flavor tensors or the physical Wilson basis.

The verification checks are:

check	result
all P11 scalar/source debt rows accounted for	true
every scalar/source row has a mass symbol	true
every source sector has a positive mass condition	true
P12 scalar/source layer passes	true

The next scalar/source stage is therefore not a proton-bound insertion. It is the construction of scalar/source flavor tensors and a chiral/Fierz Wilson basis for the BNV rows. The parallel completed current-current vector branch also remains open and should be tried before any final proton-bound claim is made.

14 P13-S: Scalar/Source Flavor Tensors and Wilson Basis

P13-S chooses a conservative symbolic flavor-tensor basis for the P12-S scalar/source branch. It does not choose numerical couplings. For every scalar/source vertex, the flavor tensor is a complex family tensor

$$\Lambda_{ab,R}^{ij}, \quad i, j = 1, 2, 3, \quad (74)$$

with pair-exchange symmetry fixed by the source channel. The selected rules are:

pair/source channel	family symmetry	components
QQ weak-singlet source	symmetric pair	6
QQ weak-triplet source	antisymmetric pair	3
$u^c u^c$ source	antisymmetric pair	3
$d^c d^c$ source	antisymmetric pair	3
distinct-field pair	general 3×3 matrix	9

The canonical all-left chiral Wilson basis chosen by P13-S is

$$O_{QQQL,S}^{prst} = \epsilon_{abc} \epsilon_{ij} \epsilon_{kl} (Q_p^{ai} Q_r^{bj}) (Q_s^{ck} L_t^l), \quad (75)$$

$$O_{QQQL,T}^{prst} = \epsilon_{abc} (\tau^A \epsilon)_{ij} (\tau^A \epsilon)_{kl} (Q_p^{ai} Q_r^{bj}) (Q_s^{ck} L_t^l), \quad (76)$$

$$O_{UUDE}^{prst} = \epsilon_{abc} (u_{p,a}^c u_{r,b}^c) (d_{s,c}^c e_t^c), \quad (77)$$

$$O_{UDDN}^{prst} = \epsilon_{abc} (u_{p,a}^c d_{r,b}^c) (d_{s,c}^c \nu_t^c). \quad (78)$$

The first two operators separate the weak-singlet and weak-triplet $QQQL$ source contractions. The last operator is a sterile-neutrino BNV basis element and cannot be interpreted as a physical proton channel without specifying the ν^c mass/seesaw or light-sterile treatment.

The generated P13-S ledger reports:

quantity	value
BNV or sterile-BNV rows processed	6
canonical Wilson basis count	4
direct source-channel rows	4
Fierz/color recoupling required rows	2
canonical family projection required rows	1
flavor tensors introduced	11
independent complex vertex components	78
physical proton bounds evaluable now	0

The two non-direct rows need explicit two-component-spinor Fierz and color recoupling matrices before their coefficients can be added to the canonical basis. In addition, the direct O_{UDDN} row needs the explicit antisymmetric projection on the two d^c family indices:

$$C_{UDDN}^{prst} \longleftarrow \Pi_{\text{asym,rs}}^{(d)} \left[\frac{\Lambda_{u^c d^c, R}^{pr} \Lambda_{d^c \nu^c, R}^{st*}}{M_{S_R}^2} \right]. \quad (79)$$

Thus P13-S has chosen the tensor and Wilson-basis bookkeeping layer, but it has not made proton bounds numerical. The remaining scalar/source tasks are the two recoupling matrices, the d -pair antisymmetrization projection, physical flavor rotations, RG evolution, hadronic matrix elements, and experimental channel limits. The parallel Branch V remains open.

15 P14-V: Vector-Branch Matching Gate

P14-V returns to the Spin(10) current-current vector branch before inserting proton limits. It does not reinterpret the P7 chiral-pair poles as adjoint vectors. Instead it starts from the P9 broken-generator maps and the P10 staged vector mass denominators. For a broken vector X , define

$$J_X^\mu = \sum_{a \in E_X} c_a \psi_{t(a)}^\dagger \bar{\sigma}^\mu \psi_{s(a)}, \quad (80)$$

where E_X is the set of sparse transition entries in the broken-generator map. The tree-level vector matching gate is

$$\mathcal{L}_{\text{eff}}^{(X)} = - \frac{g_X^2}{M_X^2} J_X^\mu J_{X,\mu}^\dagger. \quad (81)$$

The mass replacements inherited from P10 are

block	mass
$M_{(6,2,2)}$	$\sqrt{\kappa_{\text{PS}}} g_{10} v_{\text{PS}}$
M_{LQ}	$\sqrt{2/3} g_4 v_4$
M_{W_R}	$g_R v_R$
$M_{Z'}$	$\frac{1}{2} v_Y \sqrt{g_R^2 + g_{BL}^2}$

The generated P14-V ledger reports:

quantity	value
broken vector maps	32
current transition entries	80
current-pair rows	72
Pati–Salam leptoquark pair rows	36
Spin(10)/Pati–Salam off-face pair rows	24
broken $SU(2)_R$ charged-current pair rows	12
P7 rows with candidate adjoint mass blocks	0
P7 rows remaining scalar/source mass debt	18
physical proton bounds evaluable now	0
Branch V matching gate passes	true

The current-pair gates are:

gate	rows
Pati–Salam leptoquark mixed pair requiring crossing/Fierz	24
Pati–Salam leptoquark same-sector noncanonical pair	12
off-face diquark-like pair requiring crossing/Fierz	12
off-face mixed pair requiring crossing/Fierz	12
$SU(2)_R$ charged-current non-proton-seed row	12

Thus Branch V is now an explicit current-current matching gate:

$$\boxed{\text{P14-V supplies } J_X J_X^\dagger / M_X^2 \text{ rows, not proton lifetimes.}} \quad (82)$$

Before any numerical proton limit can be inserted, one must still choose a convention-fixed current-current to all-left chiral Wilson-basis map, physical flavor rotations, numerical masses and couplings, RG evolution, hadronic matrix elements, and experimental channel limits. The P13-S scalar/source branch therefore remains a parallel branch, while Branch V has its own matching gate.

16 P15-V: Current to All-Left Fierz Gate

P15-V fixes one explicit two-component Weyl Fierz convention for Branch V. The input current product is

$$(\psi_{t_1}^\dagger \bar{\sigma}^\mu \psi_{s_1})(\psi_{s_2}^\dagger \bar{\sigma}_\mu \psi_{t_2}), \quad (83)$$

where the second factor is the Hermitian-conjugate current in $J_X^\mu J_{X,\mu}^\dagger$. We use

$$(\chi^\dagger \bar{\sigma}^\mu \psi)(\eta^\dagger \bar{\sigma}_\mu \xi) = 2(\chi^\dagger \eta^\dagger)(\psi \xi). \quad (84)$$

Thus the crossed all-left bookkeeping skeleton is

$$(\psi_{s_1} \psi_{t_2})(\overline{\psi_{t_1}} \overline{\psi_{s_2}}), \quad (85)$$

with symbolic coefficient prefactor

$$-\frac{2g_X^2 c_1 c_2^*}{M_X^2}. \quad (86)$$

This is a convention gate. It is not yet a physical amplitude, since the crossed conjugate fields still need a chosen external-state convention and physical flavor basis.

The generated P15-V ledger reports:

quantity	value
P14 current-pair rows consumed	72
P15 crossed all-left rows generated	72
B, L conserving crossed skeletons	72
canonical BNV basis rows	0
physical proton bounds evaluable now	0
all rows have crossed fields	true
all rows have $B, L, B - L$ charge audit	true
Branch V all-left gate passes	true

Equivalently, under the current sparse-map pairing,

$$\boxed{\text{P15-V finds no canonical BNV proton operator in Branch V yet.}} \quad (87)$$

This does not close Branch V; it identifies the next missing datum. A physical vector-proton branch still needs Hermitian vector multiplet assembly, cross-root Clebsch phases, an external-state crossing convention, projection to a canonical BNV Wilson basis if such a projection exists, physical flavor rotations, RG running, hadronic matrix elements, and experimental channel limits. The scalar/source Branch S from P13-S remains a parallel branch and is not discarded.

17 P16-V: Physical Hermitian Vector Multiplet Assembly

P15-V used the same-map sparse-current product $J_X J_X^\dagger$ and found no canonical BNV vector operator. P16-V tests the next missing vector datum: physical Hermitian vector multiplets are assembled from charge-conjugate root maps. At the symbolic level this permits cross-root products

$$J_q^\mu J_{-q,\mu}, \quad (88)$$

where the relative physical Clebsch phase is still left as a symbolic parameter ξ . Applying the same two-component all-left Fierz convention as in P15-V gives the coefficient template

$$C_{\text{phys}} = -\frac{2g_X^2}{M_X^2} \xi c_q c_{-q}. \quad (89)$$

The canonical vector BNV skeletons searched at this gate are

$$O_{V,QQ\bar{U}\bar{E}} \sim QQ \bar{u}^c \bar{e}^c, \quad O_{V,QQ\bar{D}\bar{N}} \sim QQ \bar{d}^c \bar{\nu}^c. \quad (90)$$

These are pre-flavor and pre-hadronic skeletons, not proton-lifetime coefficients.

The generated P16-V ledger reports:

quantity	value
P15 same-map canonical BNV rows	0
physical cross-root rows	448
Pati-Salam $SU(4)_C$ leptoquark rows	144
Spin(10)/Pati-Salam off-face rows	288
broken $SU(2)_R$ charged-current rows	16
B, L conserving physical pairs	160
$B - L$ preserving BNV noncanonical pairs	144
canonical vector BNV basis candidates	144
$O_{V,QQ\bar{D}\bar{N}}$ candidates	72
$O_{V,QQ\bar{U}\bar{E}}$ candidates	72
physical proton bounds evaluable now	0
Branch V physical multiplet gate passes	true

The status by physical vector family is especially important:

block	status	rows
M_{LQ}	B, L -conserving physical pair	144
M_{W_R}	B, L -conserving physical pair	16
$M_{(6,2,2)}$	canonical vector BNV candidate	144
$M_{(6,2,2)}$	$B - L$ -preserving noncanonical BNV pair	144

(91)

Thus the canonical vector BNV candidates arise only from the off-face $M_{(6,2,2)}$ Hermitian cross-root block. The Pati–Salam leptoquark and W_R physical pairs remain B, L conserving at this bookkeeping gate.

P16-V changes the Branch-V readiness status but does not make proton limits numerical. The remaining vector-side data are

Hermitian generator normalization, cross-root Clebsch phases,
color/weak projection to the canonical BNV basis, physical flavor rotations,
numerical masses and couplings, RG factors, hadronic matrix elements, channel limits⁽⁹²⁾

The scalar/source Branch S from P13-S remains a parallel branch and is not discarded.

18 Current Status and Next Check

The P1–P16-V chain currently establishes:

1. the exact P1 charge/anomaly baseline for one $SM + \nu^c$ family;
2. the P2 charge-filtered symbolic pole components;
3. the P3 conditional residue-factorization and PSD statement;
4. the P4 Ward-identity completion ledger for broken/off-face sectors;
5. the P5 anomaly/chiral-consistency comparison across candidate branches.
6. the P6 high-energy-growth ledger for finite-vector, scalar/auxiliary, contact-only, and tower-like comparison branches.
7. the P7 conditional low-energy matching and proton-bound readiness ledger.
8. the P8 comparative scorecard across $SU(5)$, Pati–Salam, $Spin(10)$, E_6 , and superstring-derived completions.
9. the P9 $Spin(10)$ -locked sparse generator ledger and action-completion interface.
10. the P10 staged $Spin(10)$ breaking mass matrix with 33 positive broken-vector eigenvalues and one hypercharge zero mode.
11. the P11 replay gate showing that P10 supplies adjoint-vector mass denominators, while the P7 chiral-pair rows remain scalar/source mass debt.
12. the P12-S scalar/source branch that assigns symbolic scalar/source denominators to all 18 P7 chiral-pair rows, with 12 source sectors and 6 BNV or sterile-BNV rows still awaiting flavor/Wilson-basis data.

13. the P13-S scalar/source flavor/Wilson-basis layer, mapping those 6 rows to four canonical all-left chiral operators with 11 symbolic flavor tensors, 2 Fierz/color recoupling debts, and 1 direct family-projection debt.
14. the P14-V current-current vector matching gate, with 32 broken vector maps, 80 transition entries, 72 current-pair rows, and zero physical proton bounds evaluable before current/Fierz, flavor, RG, hadronic, and channel-limit inputs.
15. the P15-V all-left Fierz/crossing convention gate, mapping all 72 Branch V current-pair rows to crossed skeletons and finding 72 B, L conserving rows and zero canonical BNV rows under the present sparse-map pairing.
16. the P16-V physical Hermitian cross-root vector assembly gate, generating 448 physical cross-root rows and finding 144 canonical vector BNV skeletons in the $M_{(6,2,2)}$ off-face block before flavor, RG, hadronic, or proton-limit inputs.

The next natural stage is P17: fix the physical Hermitian generator normalization and cross-root Clebsch phase convention for the canonical vector BNV rows, then export a flavor-rotation interface. Proton limits should still not be inserted until RG and hadronic inputs are supplied.